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Code

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Console Terminal Background Jobs x
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ ↗
> # Addition
> c(2,3,5,7) + c(-2,-3,-5,8)
[1] 0 0 0 15
>
> # Recycling of shorter vector
> c(2,3,5,7) + c(8,9)
[1] 10 12 13 16
>
> # Different incompatible lengths - produces a warning
> c(2,3,5,7) + c(8,9,10)
[1] 10 12 15 15
Warning message:
In c(2, 3, 5, 7) + c(8, 9, 10) :
  longer object length is not a multiple of shorter object length
>
> # Subtraction
> c(2,3,5,7) - c(-2,-3,-5,8)
[1] 4 6 10 -1
>
> c(12,13,15,17) - c(8,9)
[1] 4 4 7 8
>
> # Different incompatible lengths - produces a warning
> c(12,13,15,17) - c(8,9,10)
[1] 4 4 5 9
Warning message:
In c(12, 13, 15, 17) - c(8, 9, 10) :
  longer object length is not a multiple of shorter object length
>
> # Multiplication
> c(2,3,5,7) * c(-2,-3,-5,8)
[1] -4 -9 -25 56
>
> c(2,3,5,7) * c(8,9)
[1] 16 27 40 63
>
> # Different incompatible lengths - produces a warning
> c(2,3,5,7) * c(8,9,10)
[1] 16 27 50 56
Warning message:
In c(2, 3, 5, 7) * c(8, 9, 10) :
  longer object length is not a multiple of shorter object length
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Console	Terminal	Background Jobs	Console	Terminal	Background Jobs
R 4.6.1 · ~/Documents/R Programming/NPTEL/ ↗			R 4.6.1 · ~/Documents/R Programming/NPTEL/ ↗		
<pre> > > # Division > c(24,20,8,16) / c(3,4,2,8) [1] 8 5 4 2 > > c(24,20,8,16) / c(4,2) [1] 6 10 2 8 > > # Different incompatible lengths - produces a warning > c(24,20,8,16) / c(4,2,8) [1] 6 10 1 4 Warning message: In c(24, 20, 8, 16)/c(4, 2, 8) : longer object length is not a multiple of shorter object length > > > # --- Lecture 7: Assignment and Data Types --- > > x <- 20 > x = 20 > > y = x * 2 > z = x + y > > # Numbers and characters > x <- 20 > x = 20 > > x = "apple" > x <- "apple1" > x = 'apple' > x <- 'apple' > > # Numeric / character checks > x = 20 > is.numeric(x) [1] TRUE > is.character(x) [1] FALSE > > y = "apple" > is.character(y) [1] TRUE </pre>			<pre> > is.numeric(y) [1] FALSE > > # Converting number to character > x = 20 > is.numeric(x) [1] TRUE > > y = as.character(x) > is.numeric(y) [1] FALSE > is.character(y) [1] TRUE > y [1] "20" > > # Converting character to number > y = "apple" > is.numeric(y) [1] FALSE > is.character(y) [1] TRUE > > z = as.numeric(y) Warning message: NAs introduced by coercion > is.numeric(z) [1] TRUE > is.character(z) [1] FALSE > z [1] NA > > # Comments > # mu is the mean > # x <- 20 is treated as a comment only > > # Case sensitivity > X <- 20 > x <- 20 > > # Combining values into a vector > c(1,2,3,4,5) [1] 1 2 3 4 5 > </pre>		

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 R 4.6.1 · ~/Documents/R Programming/NPTEL/		
<pre>> # mode() > x = 6 > x [1] 6 > mode(x) [1] "numeric" > > y = "apple" > y [1] "apple" > mode(y) [1] "character" > > # storage.mode() > x = 6 > storage.mode(x) [1] "double" > > x = TRUE > storage.mode(x) [1] "logical" > > x = "apple" > storage.mode(x) [1] "character" > > # Infinity > 3/0 [1] Inf > 5 + Inf [1] Inf > > x = 5 + Inf > is.finite(x) [1] FALSE > is.infinite(x) [1] TRUE > > > # --- Lecture 8: R as a Calculator --- > > 2 + 3 [1] 5 > 2 * 3 [1] 6</pre>	<pre>> 2 - 3 [1] -1 > 3 / 2 [1] 1.5 > 2 * 3 - 4 + 5 / 6 [1] 2.833333 > > # BODMAS > (2+3)*5 + 5 - 10 [1] 20 > (((2+3)*5 + 5) - 10) / 2 [1] 10 > > # Spaces do not affect calculations > 2+5 [1] 7 > 2 + 5 [1] 7 > 2 + 5 [1] 7 > 2 +5 [1] 7 > > # Scalar and vector operations > c(2,3,5,7) + 10 [1] 12 13 15 17 > c(12,13,15,17) - 10 [1] 2 3 5 7 > c(2,3,5,7) * 10 [1] 20 30 50 70 > c(12,13,15,17) / 10 [1] 1.2 1.3 1.5 1.7 > </pre>	